

Release or Hold?

An Engineering Ethics Dilemma at the Aktobe Reservoir
During the 2024 Floods

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THE QUESTION

When floodwater arrives faster than a dam can pass it, engineers must choose who bears the risk — the structure, or the people living below it.

01 What the engineers were facing

The engineers were not prepared. The danger was never forecast.

Snow was not the problem. In spring 2024 the snow in the reservoir's catchment held 97 % of the normal water. In 2023 it held 130 % — and nothing happened. On 4 March Kazhydromet even reported 16–68 % less snow water than normal in the Ilek basin.

So the problem was not *how much* water came.

It was *how fast* it came.

Kazhydromet, spring forecast 04.03.2024 and annual snow bulletin 2024

FIVE CAUSES THAT CAME TOGETHER

1

Sudden thaw, 24 March

Temperatures rose rapidly. Instead of melting gradually over weeks, the snow melted almost at once.

“On 24 March it became warm suddenly and the snow began to melt.” — akim of Kobda district

2

Frozen ground

The soil was wet before winter and then froze deeply. Meltwater could not soak in; almost all of it flowed over the surface.

This is what turns a normal snowpack into an extreme inflow.

3

Ice crust, up to 6.6 cm

Kazhydromet recorded an ice crust on the soil on 4 March, including in the Ilek basin.

The crust acted as a barrier, directing water into rivers rather than allowing it to soak into the ground.

4

Rain on melting snow

Two days of rain fell while the snow was already melting.

Kazhydromet had described this exact scenario three weeks earlier: fast warming plus heavy March rain can create a snowmelt-and-rain flood.

5

Runoff from the steppe

Water did not stay in the river channels. It moved as a wide front across the open steppe.

“On 26 March meltwater from the steppe began to flood the city.” — akim of Aktobe

WHAT IT LED TO

1,700 m³/s

Peak inflow — operator's account

~12 hours

For the reservoir to fill

200 million m³

Inflow 27–29 March = 80 % of design volume

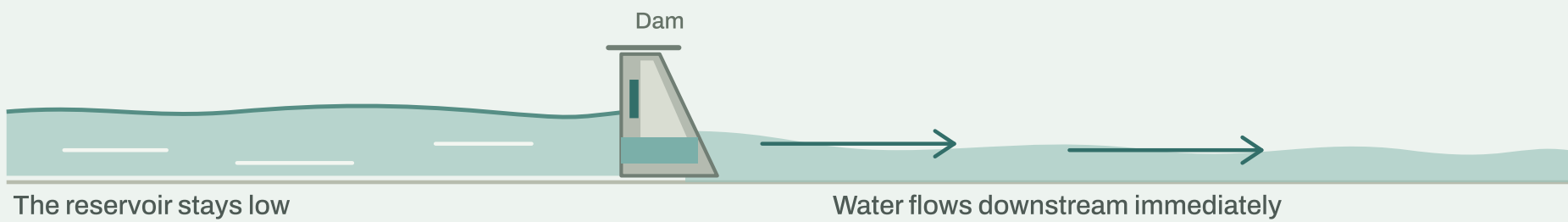
68 %

Already full before it started

02 Two options, difficult consequences

RELEASE

Open the gates now. Allow the floodwater to pass through at full speed.



✓ PROTECTS — THE DAM

- The water level never goes above the **safe limit**.
- The dam stays inside its **safety margin**.
- No risk that the dam **breaks** and floods the whole city.

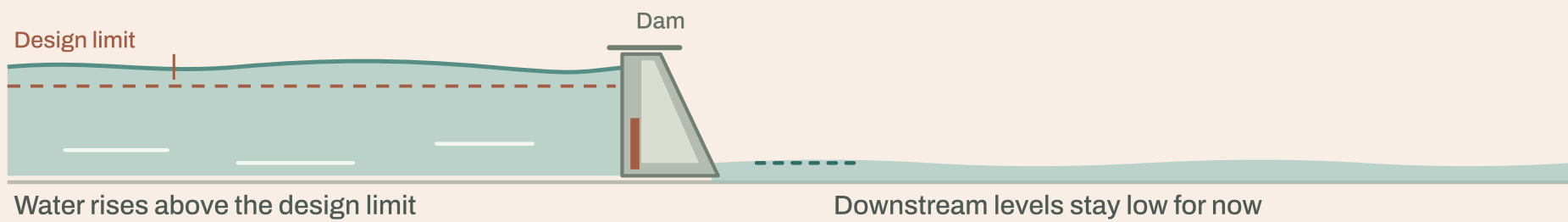
✗ COSTS — THE PEOPLE BELOW

- The water reaches the houses **at once**.
- People are still at home — **no warning** has been given.
- **No buses and no shelters** are ready yet.
- People escape **during** the flood, not before it.

In the same region, a reservoir manager was later sent to court for not releasing water in time.

HOLD

Keep the gates almost closed. Store the incoming floodwater in the reservoir.



✓ PROTECTS — THE PEOPLE BELOW

- The settlements stay **dry**.
- Homes and property are **not touched**.
- The hours gained can be used to **warn and evacuate** people.

✗ COSTS — THE DAM

- The water rises above the **design level**.
- The dam is 36 years old and was never rebuilt.
- If it breaks, the water is **no longer controlled**.
- An uncontrolled wave would reach the **city itself**.

One week later the Magadzhnan dam upstream did break. Three settlements were flooded.

03 The decision — 28 March 2024



The response combined both approaches. To enable the evacuation of residents downstream, the operator — *together with local authorities* — deliberately stored **15 million m³ above the reservoir's design capacity**, accepting extra load on the dam to buy hours. Only then was discharge raised to **1,000 m³/s**.

“As a result of the decision to release, the garden communities of Aktobe were flooded.”

Press service of the Prime Minister of Kazakhstan, 29 March 2024

FROM DENIAL TO EVACUATION ORDER

23 h 25 min

27 Mar, 22:38 — “This audio is false. The reservoir is only 68 % full.”

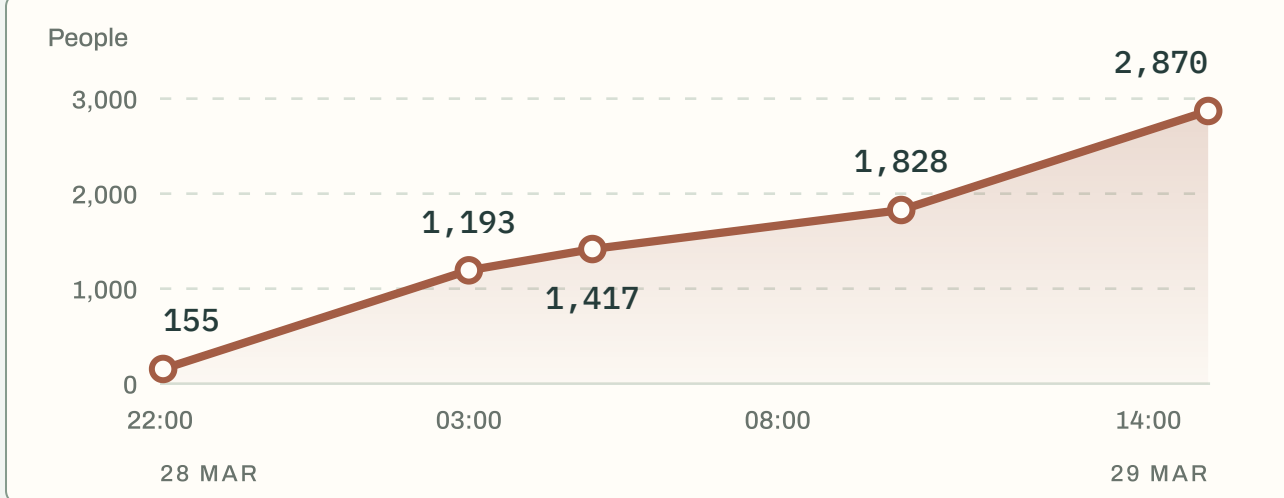
28 Mar, 22:03 — “We are forced to release water. Please evacuate.”
Emergency evacuation starts.

Night 28 → 29 Mar — the gates are open, the water is released.

Timestamps from the official akimat channel. The exact hour the gate was opened has never been published.

04 What followed

PEOPLE IN EVACUATION CENTRES, 28–29 MARCH



689
garden homes flooded

2,870
evacuated within a day

0
lives lost

540 mln m³
passed through — 2.2 reservoir volumes



Map: the reservoir, the dam and the Ilek river below it



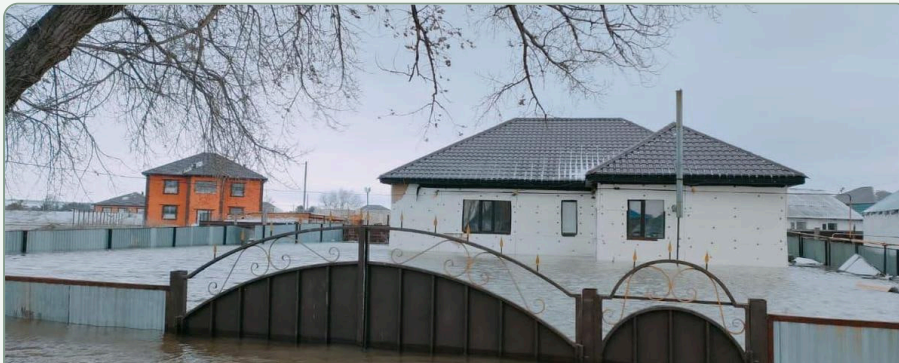
Google Maps satellite: the dam and its spillway



Map of the flooded areas of Aktobe



A whole settlement under water



Houses flooded to window level



Sandbag barriers — 157,278 sacks laid in the region

05 Lessons for future flood preparedness

1 · Prepare before the flood season

- Empty the reservoirs **before** the flood season.
- Break the river ice early so it cannot form jams.
- Before 2025 the reservoirs were pre-empted to free **382 million m³**.

2 · Maintain dams and equipment

- Inspect and repair gates, cranes and dams **every year** — not after a disaster.
- In 2024 the crane needed to open the second gate **failed at the worst moment**.
- 537 structures in Kazakhstan still need repair; 322 threaten 1.5 million people.

3 · Prepare people for an emergency

- Plan the evacuation **before** the water comes: routes, buses, shelters.
- Warn people early and clearly, so the state and the population **understand each other**.
- A warning works only if people **believe** it and can act on it.

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